

Note 5.6 Substitution and Area Between Curves

The Substitution Formula

- Theorem 7--Substitution in Definite Integrals:**

If g' is continuous on the interval $[a, b]$ and f is continuous on the range of $g(x) = u$, then

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

Proof:

Let F be an antiderivative of f , then

$$\begin{aligned}\int_a^b f(g(x))g'(x) dx &= F(g(x)) \Big|_{x=a}^{x=b} \\ &= F(g(b)) - F(g(a)) \\ &= F(u) \Big|_{u=g(a)}^{u=g(b)} \\ &= \int_{g(a)}^{g(b)} f(u) du\end{aligned}$$

Definite Integrals of Symmetric Functions

- Theorem 8:**

Let f be continuous at a symmetric interval $[-a, a]$.

- If f is even, then $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$.
- If f is odd, then $\int_{-a}^a f(x) dx = 0$.

Proof:

- For the first part,

$$\begin{aligned}\int_{-a}^a f(x) dx &= \int_0^a f(x) dx + \int_{-a}^0 f(x) dx \\ &= \int_0^a f(x) dx - \int_0^{-a} f(-x) dx \\ &= \int_0^a f(x) dx - \int_0^a f(u) (-du) \\ &= \int_0^a f(x) dx + \int_0^a f(u) du \\ &= 2 \int_0^a f(x) dx\end{aligned}$$

(At the last step, if we substitute back $-x$, then the upper limit need to change as well.)

$$\begin{aligned}
\int_{-a}^a f(x) dx &= -\int_0^{-a} f(x) dx + \int_0^a f(x) dx \\
&= \int_0^{-a} f(-x) dx + \int_0^a f(x) dx \\
&= \int_0^a f(u) (-du) + \int_0^a f(x) dx \\
&= -\int_0^a f(u) du + \int_0^a f(x) dx \\
&= 0
\end{aligned}$$

Areas Between Curves

- Definition:**

If f and g are continuous with $f(x) \geq g(x)$ on $[a, b]$, then the area of the region between the curves $y = f(x)$ and $y = g(x)$ from a to b is the integral of $(f - g)$ from a to b :

$$A = \int_a^b [f(x) - g(x)] dx$$

Proof:

We can approximate the region with n vertical rectangles by dividing it into n subintervals. A partition is $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$. Then the area of the k th rectangle is

$$A_k = [f(c_k) - g(c_k)] \cdot \Delta x_k$$

then the area of the whole region is

$$A = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n [f(c_k) - g(c_k)] \Delta x_k = \int_a^b [f(x) - g(x)] dx$$

Integration with Respect to y

- Definition:**

If a region's bounding curves are described by functions of y , the approximating rectangles are horizontal and the formula has y instead of x .

$$A = \int_c^d [f(y) - g(y)] dy$$

and in the formula f is always on the right side of g to ensure the value is nonnegative.

Exercises

- P350 3,27,45